

Dineth Perera

Electrical & Electronic Engineering Undergraduate | Remote-Sensing Computer Vision | Signal Processing

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Professional Summary

Final-year B.Sc. Engineering undergraduate in Electrical and Electronic Engineering at the University of Peradeniya (GPA: 3.74/4.00), with research interests in computer vision, remote sensing, Earth observation, signal processing, and machine learning. Current research focuses on remote-sensing change detection, visual state-space models, boundary-aware representation learning, and multi-temporal Earth observation imagery. Experienced in developing, training, benchmarking, and analysing deep-learning models across remote-sensing datasets.

Currently seeking **research internship and visiting research opportunities for 2027** in computer vision, remote sensing, Earth observation, machine learning, signal processing, and related areas.

Education

B.Sc. Engineering (Hons) in Electrical and Electronic Engineering

2023–Present

University of Peradeniya, Sri Lanka

- Current GPA: **3.74/4.00**.
- Academic interests include computer vision, remote sensing, signal processing, control systems, machine learning, and embedded systems.

G.C.E. Advanced Level – Physical Science Stream

2021

- National Rank: **68 out of 33,745** candidates.
- District Rank: **4**.
- Z-score: **2.62**.

Publications and Pre Prints

MambaRefine-CD: MambaVision with Region–Boundary Temporal Refinement

Moratuwa Engineering Research Conference (MERCon), 2026 – Accepted, Presented, and Published in IEEE Xplore
Dineth Perera, Thaarig Firdous, Oshadha Samarakoon, Roshan Godaliyadda, Parakrama Ekanayake, Vijitha Herath.

[IEEE Xplore](#) [Code](#)

- Proposed a binary remote-sensing change-detection framework based on a shared **MambaVision-S** encoder with explicit region–boundary temporal refinement.
- Introduced **Differential Region–Boundary Interaction (D-RBI)**, which constructs temporal evidence from paired features, absolute differences, and signed differences, and separates this evidence into region-aware and Sobel-conditioned boundary streams.
- Integrated **CRAM-lite** region modulation, an **Adaptive Receptive-Field FPN (ARF-FPN)**, and a bounded residual refinement head for controlled boundary correction.
- Evaluated the framework on **DSIFN-CD** and **WHU-CD**, achieving average F1/IoU scores of **95.67/91.71%** and **95.34/91.10%**, respectively, under the verified evaluation protocols used in the study.

A Controlled Benchmark of Visual State-Space Backbones with Domain-Shift and Boundary Analysis for Remote-Sensing Segmentation

IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2026 – Accepted and Presented

Nichula Wasalathilaka, **Dineth Perera**, Oshadha Samarakoon, Buddhi Wijenayake, Roshan Godaliyadda, Vijitha Herath, Parakrama Ekanayake.

[arXiv:2604.18721](#) [Code](#)

- Developed a strictly controlled benchmark of **VMamba**, **MambaVision**, and **Spatial-Mamba** backbones for remote-sensing semantic segmentation while holding the decoder, feature interface, training schedule, preprocessing, and evaluation protocol fixed.
- Evaluated visual state-space models on **LoveDA** and **ISPRS Potsdam**, including in-domain segmentation, source-only cross-domain generalization, boundary behaviour, computational efficiency, and model scaling.
- The study showed that visual state-space models provide favourable accuracy–efficiency trade-offs under the controlled setup, while **boundary delineation remains a major failure mode under domain shift** and encoder scaling alone provides only modest gains.

Research in Progress

ORBIT-Mamba: Operator-Conditioned Region–Boundary Temporal Specialization for Remote Sensing Change Detection

Manuscript in final preparation for submission to IEEE Transactions on Geoscience and Remote Sensing (TGRS)

- Developing an **operator-conditioned change-detection framework** based on the hypothesis that semantic Region recognition and structural Boundary localization benefit from different forms of bi-temporal evidence.
- Uses a shared **VMamba-S** encoder followed by independently selected Region and Boundary temporal operators and separate **Differential Region–Boundary Modulation (D-RBM)** pathways.
- Evaluated across **DSIFN-CD, SYSU-CD, WHU-CD, and LEVIR-CD+**, analysing both semantic change-detection performance and boundary quality.

Multi-Hazard Remote Sensing Dataset for Disaster Change Analysis

Dataset Development – Ongoing

- Developing a multi-hazard Earth-observation dataset covering **wildfire, tsunami, flood, and landslide** events for disaster-related remote-sensing change analysis.
- Curating and organizing multi-temporal satellite imagery to support research on disaster evolution, affected-area localization, and machine-learning-based change detection.
- Designing a consistent dataset organization and metadata structure for bi-temporal and, where suitable, multi-temporal disaster-analysis experiments.

Engineering Projects

Real-Time Elephant Detection Using Infrasound Analysis and Machine Learning

- Developing an infrasound-based elephant-detection approach using low-frequency acoustic signals approximately within the 5–35 Hz range.
- Investigating signal acquisition, preprocessing, noise suppression, spectral feature extraction, and machine-learning classification.

Technical Skills

- **Programming & Scientific Computing:** Python, C/C++, MATLAB, LaTeX, Git, Linux.
- **Deep Learning & Computer Vision:** PyTorch, TensorFlow, OpenCV; CNNs, Vision Transformers, Visual State-Space/Mamba models, semantic segmentation, binary change detection, representation learning, model training.
- **Remote Sensing & Earth Observation:** Bi-temporal and multi-temporal image analysis, remote-sensing change detection, Sentinel-2 imagery, dataset preparation, image/mask processing.
- **Machine Learning & Experimental Research:** Dataset preparation, model implementation, controlled ablation studies, performance analysis, hyperparameter experimentation, reproducible evaluation, and scientific visualization.
- **Signal Processing:** FFT/PSD analysis, digital filtering, spectral analysis, frequency-domain feature extraction, low-frequency signal analysis, and infrasound processing.
- **Embedded Systems & Edge AI:** Arduino/AVR, ESP32, Raspberry Pi, NVIDIA Jetson, sensor interfacing, embedded data acquisition.

Honors and Awards

Dharmaraja College, Kandy

- **P.D.S. Kularathne Memorial Prize and N.B.M. Mediweka Memorial Prize** – awarded for the best student selected to the Faculty of Engineering in 2021.
- **S.J. Munasinghe Memorial Prize** – awarded to the student obtaining the highest Z-score at the 2021 G.C.E. Advanced Level Examination.

Extracurricular Activities

Violinist – Dharmaraja College, Kandy

- Recipient of **multiple national-level awards** in violin performance.

Cricket Player – Dharmaraja College, Kandy / Provincial Team - Central

- Represented **school and provincial-level cricket team** .

Central Link Toastmasters Club

2023–2026

- Member, developed public speaking, technical communication, structured presentation, and professional communication skills through club participation.

Academic References

Prof. Roshan Godaliyadda

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Prof. Parakrama Ekanayake

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